Tanmay Bhuskute

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SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with production experience building backend systems, infrastructure automation, and distributed platforms in Python, Go, C/C++, and AWS. Current research applies AI/ML to real-time system telemetry and automated event analysis.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: C/C++, Go, Python, Java, JavaScript, TypeScript, SQL.
Systems & APIs: Linux, REST APIs, gRPC, TCP/socket programming, event-driven architecture, concurrent programming.
Cloud & DevOps: AWS (EC2, S3, SQS, Lambda, CloudWatch), Docker, Kubernetes, Terraform, GitLab CI/CD, Git, Linux, Bash.
Observability & AI: CloudWatch, logging/telemetry pipelines, anomaly detection, Gemini, Vertex AI, PyTorch, scikit-learn.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Built provider-agnostic event-analysis pipeline processing real-time Android telemetry streams, decoupling video understanding from device automation for automated bug reproduction across 53 runs on 20+ apps.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) via memory-augmented inference over per-event VLM replay, preserving task-completion fidelity.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** infrastructure workflows with Python (Boto3) and REST APIs, replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments.
- Executed **44 production environment upgrades** in 4 months, diagnosing live infrastructure and service issues across Linux, EC2, and distributed databases using CloudWatch log correlation.
- Designed **Ansible-based Blue-Green deployment** framework improving fault tolerance, reducing production upgrade downtime by **50%**.
- Migrated **40+ production environments** to Kubernetes using ArgoCD, Helm, and GitLab CI/CD, improving system resiliency and scalability.
- Served as Quality Coach and SonarQube Specialist, increasing automated code coverage from **35% to 78%**; mentored **5 software engineers** through code review and onboarding.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework for infrastructure testing, catching **95%** of misconfigurations pre-production.

Software Developer Intern —- Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management, automated manual processes, and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [Python, PySpark, FastAPI, Docker, PostgreSQL] Jul 2026

- Built distributed data pipeline processing **2.3M-row** streaming transportation dataset across **263 zones**; served real-time predictions via REST API with PostgreSQL and Redis caching, cutting RMSE **43%** over baseline.

PolicyPilot – Medical Benefit Policy Intelligence Portal [Node.js, TypeScript, Docker, MCP] May 2026

- Designed **17-tool** backend API server with Zod schema validation at every data boundary, exposing structured tool-calling to external agents and eliminating a class of runtime type errors.
- Engineered confidence-scoring system for automated anomaly and inconsistency detection across ingested documents, driving human-review triage decisions.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).